

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

PHYSICS | SSC-I (9th Grade) | Units 1-6 and 9 | Marking Report

Exam: exam-08-ch1-6-9 (Version 1)

Candidate: Seemab

Date: 2 July 2026

Total Marks: 65

Deep-check verified score: arithmetic slip confirmed on Q.2(vi)(or). 120/6 written as 60 instead of 20. Setup correct (2/3). Final score corrected from an initial reading of 62/65 to the verified **61/65 (93.8%)**.

61 / 65 93.8% A+

Section A: 12/12 | Section B: 30.5/33 | Section C: 18.5/20 | MCQ Streak: 10 consecutive perfect

SECTION A: MCQS

12/12

100% (Perfect)

SECTION B: SHORT QUESTIONS

30.5/33

92.4% (3 deductions)

SECTION C: LONG QUESTIONS

18.5/20

92.5% (2 deductions)

SECTION A: Multiple Choice Questions (12 x 1 = 12 Marks)

#	Question (abbreviated)	Seemab	Correct	Result
1	Prefix nano represents a multiplier of:	B	B (10 to the power of -9)	CORRECT
2	Screw gauge pitch 1mm, 100 divisions. Least count:	B	B (0.01 mm)	CORRECT
3	Area under a velocity-time graph represents:	B	B (Distance covered)	CORRECT
4	Car: rest to 30 m/s in 6 s. Acceleration:	A	A (5 m/s squared)	CORRECT
5	SI unit of impulse is:	B	B (N-s)	CORRECT
6	Body of 2 kg at 5 m/s. Momentum:	A (corrected from B)	A (10 kg-m/s)	CORRECT
7	Centripetal force on a body in a circle is directed:	C	C (Towards the centre)	CORRECT
8	Smallest friction for same pair of surfaces:	C	C (Rolling friction)	CORRECT
9	Force 200 N on 0.5 m squared. Pressure:	B	B (400 Pa)	CORRECT
10	Work done zero when angle between force and displacement:	C	C (90 degrees)	CORRECT
11	2 kg body at 3 m/s. Kinetic energy:	B	B (9 J)	CORRECT
12	Concise mathematical statement of a natural relationship:	C	C (Law)	CORRECT

MCQ Score: 12/12 (Perfect). **10th consecutive perfect MCQ score** Q.6 had an initial B marked then correctly changed to A before submission.

SECTION B: Short Answer Questions (11 x 3 = 33 Marks)

All 11 parts attempted. OR variant chosen for Q.2(i), Q.2(ii), Q.2(v), Q.2(vi), Q.2(x).

Q.2(i) OR: Significant Figures

Chose: OR alternative

3/3

- (a) 0.0067 m: stated 2 significant figures, correctly noting leading zeros are not significant.
 (b) 30500 J: stated 3 significant figures (trailing zeros without decimal point not counted).
 (c) 4.5678 rounded to 3 sig figs: gave 4.57, correct (7 rounds up the 6).

Q.2(ii) OR: SI Prefixes and Unit Conversion

Chose: OR alternative

3/3

- (i) 10 to the power 6 = mega (M); (ii) 10 to the power -3 = milli (m); (iii) 10 to the power -9 = nano (n). All three correct (2 marks).
 (b) 5 km = 5 x 10 cubed m = 5000 m. Correct conversion (1 mark).

Q.2(iii) MAIN: Distance, Displacement, Average Speed

Chose: Main question

2.5/3

- (i) Total distance = 300 m + 100 m = 400 m. Correct (1 mark).
(ii) Net displacement written as 300 m - 100 m = 200 m. Magnitude correct but direction (East) not stated. Displacement is a vector and requires both magnitude and direction for full credit. Deduction: -0.5.
 (iii) Total time = 20 s + 10 s = 30 s; average speed = 400/30 = 13.3 m/s. Correct (1 mark).

Q.2(iv) MAIN: Newton's Second Law and Mass Calculation

Chose: Main question

2/3

Newton's second law stated correctly in words: "The acceleration of an object is directly proportional to the magnitude of the net force in the same direction, and inversely proportional to the mass of an object." (2 marks for verbal statement)

Scan observation: the proportionality steps read "a proportional to m" then "a proportional to 1/mass". The first line has the proportionality to m (mass directly) rather than to F (force). This is a miswrite in the working steps; the verbal law statement is correct and $F = ma$ was derived. No additional deduction beyond the missing numerical result.

Numerical calculation not completed. The question asks to calculate the mass given $F = 24 \text{ N}$ and $a = 3 \text{ m/s squared}$. Required: $m = F/a = 24/3 = 8 \text{ kg}$. This calculation was entirely absent. Deduction: -1.

Q.2(v) OR: Change in Momentum and Impulse (Cricket Ball)

Chose: OR alternative

3/3

Law stated: "impulse = change in momentum" (1 mark).

$\Delta p = m(v - u) = 0.15 \times (-20 - 20) = 0.15 \times (-40) = -6 \text{ kg}\cdot\text{m/s}$; magnitude $6 \text{ kg}\cdot\text{m/s}$. Direction-aware calculation with self-correction shown. (2 marks)

Impulse = change in momentum = $6 \text{ N}\cdot\text{s}$. Stated and correct.

Q.2(vi) OR: Principle of Moments

Chose: OR alternative

2/3

Principle of moments stated correctly: when a body is in equilibrium, the clockwise moment equals the anticlockwise moment about the same pivot. (1 mark)

Setup correct: Anticlockwise = Clockwise; $4 \text{ N} \times 30 \text{ cm} = 6 \times d$; $120 \text{ N}\cdot\text{cm} = 6 \times d$. (1 mark)

Arithmetic slip: 120 divided by 6 was written as 60 (not 20). The scan shows a crossed-out fraction with "60" written above it, and the final line reads " $60 = d$ ". The correct answer is $d = 20 \text{ cm}$. The answer 60 cm is physically impossible (a metre rule pivoted at centre has a maximum arm of 50 cm per side). No sanity check was applied. Deduction: -1.

Q.2(vii) MAIN: Centripetal Force

Chose: Main question

3/3

Centripetal force defined: "the net force directed towards the centre of a circle that keeps a body moving along a circular path." (1 mark)

Formula: $F_c = mv^2 / r$ stated.

Calculation: $F_c = 0.5 \times (4)^2 / 1 = 0.5 \times 16 / 1 = 8 \text{ N}$. Correct. (2 marks)

Q.2(viii) MAIN: Hooke's Law and Spring Constant

Chose: Main question

3/3

Hooke's law stated: within the elastic limit, the extension produced in a spring is directly proportional to the applied force ($F = kx$). (1 mark)

Spring constant defined: "the force needed to produce unit extension"; SI unit N/m . (1 mark)

$k = F/x = 12 / 0.04 = 300 \text{ N/m}$ (correctly converted 4 cm to 0.04 m). (1 mark)

Q.2(ix) MAIN: Pressure Definition and Box on Floor

Chose: Main question

3/3

Pressure defined: "the force acting normally (perpendicularly) per unit area of a surface"; $P = F/A$; SI unit pascal (Pa), $1 \text{ Pa} = 1 \text{ N/m squared}$. (2 marks)

$P = F/A = 600 \text{ N} / 1.5 \text{ m squared} = 400 \text{ Pa}$. Correct. Stated both 400 Pa and 400 N/m squared showing understanding of unit equivalence. (1 mark)

Q.2(x) OR: Power Definition and Pump Output

Chose: OR alternative

3/3

Power defined: "rate of doing work (workdone per unit time)"; $P = W/t$; SI unit watt (W), where $1 \text{ W} = 1 \text{ J/s}$. (2 marks)

$P = W/t = 6000 \text{ J} / 30 \text{ s} = 200 \text{ W}$. Correct. (1 mark)

Q.2(xi) MAIN: Steps of the Scientific Method

Chose: Main question

3/3

Listed five steps in correct order: (1) Observation of a problem or phenomenon; (2) Forming a hypothesis; (3) Performing experiments to test the hypothesis; (4) Analysing the results and drawing a conclusion; (5) If confirmed, forming a theory and law. All five steps present and in correct order. (3 marks)

Section B Total: 30.5 / 33 (92.4%)**SECTION C: Long Answer Questions (4 x 5 = 20 Marks)**

All 4 long questions attempted. Main chosen for Q.3(i) and Q.3(ii); OR chosen for Q.3(iii) and Q.3(iv).

Q.3(i) MAIN: Acceleration, Distance, v-t Graph Interpretation

Chose: Main question

4/5

Acceleration defined: "rate of change of velocity of a body"; formula $a = (v_f - v_i)/t$; SI unit m/s squared stated. (1 mark)

(a) $a = (30 - 10)/8 = 20/8 = 2.5 \text{ m/s squared}$. Correct. (1 mark)

(b) Distance: $S = (v_i + v_f)/2 \times t = (10 + 30)/2 \times 8 = 20 \times 8 = 160 \text{ m}$. Correct method and answer. (2 marks)

(c) "State what the area under its velocity-time graph represents" was not answered at all. The required statement: "the area under a velocity-time graph equals the distance covered by the body." Deduction: -1.

This sub-part needed only a single sentence of theory. Re-reading the full question text before moving on would have caught it. Part of the recurring stopping-before-all-sub-parts pattern (8th consecutive exam).

Momentum defined: "the quantity of motion of a body, equal to mass x velocity; $p = mv$; it is a vector quantity." (1 mark)

Law of conservation of momentum: the verbal statement ("in an isolated system, the total momentum before an interaction equals the total momentum after") was not written. The conservation principle was used correctly in the calculation. Lenient marking applied: -0.5. Note that strict FBISE marking gives -1 since the answer key allocates a full mark specifically for the word statement of the law.

Before collision: $p = m_1v_1 + m_2v_2 = 1500 \times 12 + 1000 \times 0 = 18,000 \text{ kg}\cdot\text{m/s}$. (1 mark)

After (locked together): $18,000 = (1500 + 1000) \times v$; $18,000 = 2500v$; $v = 7.2 \text{ m/s}$. Correct. (2 marks)

Q.3(iii) OR: Derivation of $P = \rho \times g \times h$ and Total Pressure

Chose: OR alternative

5/5

Full derivation: liquid column of height h , base area A ; Volume $= Ah$; mass $= \rho \times A \times h$; weight (force on base) $= \rho \times A \times h \times g$; Pressure $= \text{force/area} = \rho \times g \times h$. All steps logically ordered and clearly written. (3 marks)

Liquid pressure $= \rho \times g \times h = 1000 \times 10 \times 5 = 50,000 \text{ Pa}$. (1 mark)

Total pressure $= \text{atmospheric} + \text{liquid} = 1.0 \times 10 \text{ to the power } 5 + 0.5 \times 10 \text{ to the power } 5 = 1.5 \times 10 \text{ to the power } 5 \text{ Pa}$. Correct and clearly laid out. (1 mark)

Q.3(iv) OR: Law of Conservation of Energy (Ball Drop)

Chose: OR alternative

5/5

Law of conservation of energy stated in full: "energy can neither be created nor destroyed; it only changes from one form to another, so the total energy of a system stays constant." Textbook-precise statement. (1 mark)

(a) PE at top $= mgh = 3 \times 10 \times 5 = 150 \text{ J}$. Correct. (1.5 marks)

(b) KE just before hitting ground $= 150 \text{ J}$ (all PE converted to KE by conservation). Stated explicitly and correctly. (1 mark)

(c) $0.5 \times m \times v^2 = 150$; rearranged: $v^2 = (150 \times 2) / 3 = 100$; $v = 10 \text{ m/s}$. Correct algebra throughout. (1.5 marks)

Section C Total: 18.5 / 20 (92.5%)

Complete Deduction Log (reconciles to headline score)

Question	Awarded	Lost	Reason
Q.2(iii): displacement direction	2.5/3	-0.5	Displacement magnitude 200 m correct; direction (East) not stated; displacement is a vector requiring both magnitude and direction
Q.2(iv): mass calculation missing	2/3	-1	Newton's 2nd law stated correctly in words (2/2); numerical result $m = F/a = 24/3 = 8 \text{ kg}$ entirely absent
Q.2(vi): arithmetic slip on 120/6	2/3	-1	Setup correct (statement 1/1, equation 1/1); 120 divided by 6 written as 60 instead of 20; 60 cm is physically impossible on a half-metre arm; no sanity check applied
Q.3(i)(c): v-t graph interpretation	4/5	-1	Sub-part (c) "state what area under v-t graph represents" not answered; only parts (a) and (b) completed; one sentence was needed
Q.3(iii): conservation law not in words	4.5/5	-0.5	Conservation of momentum used correctly in calculation but verbal statement not written; lenient -0.5 applied (strict FBISE: -1)
Total deductions		-4	$65 - 4 = 61/65$ confirmed

Strengths on This Paper

- Perfect 12/12 MCQ (10th consecutive); Q.6 self-corrected from B to A showing good checking instinct
- Significant figures, prefix table, and unit conversion all flawless
- Impulse calculation fully direction-aware: $v = -20 \text{ m/s}$ correctly taken for return direction, $\Delta p = -6 \text{ kg}\cdot\text{m/s}$, magnitude $6 \text{ N}\cdot\text{s}$
- Hooke's law: correct 4 cm to 0.04 m conversion, $k = 300 \text{ N/m}$ precise
- Full derivation of $P = \rho \times g \times h$ with all steps logically ordered (5/5)
- Conservation of energy: law stated with full textbook precision; all three sub-parts answered; algebra clean from $\text{KE} = 150 \text{ J}$ to $v = 10 \text{ m/s}$ (5/5)
- Scientific method: all 5 steps in correct order without any prompt

Areas to Revise

- Arithmetic sanity checks:** Q.2(vi): 120 divided by 6 was computed as 60. The answer 60 cm is physically impossible on a half-metre arm. Always check whether the numerical answer makes physical sense before writing it down
- Direction for vector quantities:** displacement, momentum, force answers must include direction. "200 m" is incomplete; "200 m East" is the correct form
- Complete every sub-part before moving on (8th consecutive exam):** Q.3(i)(c) needed one sentence and was skipped entirely. Q.3(ii) needed one verbal sentence for the law. Combined cost: 1.5 marks. Re-read the full question and tick each lettered part before moving to the next question
- Law statements in words before the formula:** FBISE awards a separate mark for verbal law statements. Showing the algebra without first writing the law cost half a mark (a full mark under strict marking)
- Numerical answers when a question says "calculate":** Q.2(iv) stated Newton's second law perfectly but did not compute $m = 8 \text{ kg}$

Physics Exam Comparison

Exam	Date	Score	%	MCQ	Sec B	Sec C	Grade
exam-01-ch1-2	7 Apr 2026	63.5/65	97.7%	12/12	32.5/33	19/20	A+
exam-02-ch3-4-9	10 Apr 2026	62/65	95.4%	12/12	30.5/33	19.5/20	A+
exam-03-ch5	18 May 2026	65/65	100%	12/12	33/33	20/20	A+
exam-04-ch6	22 Jun 2026	55.5/65	85.4%	12/12	29/33	14.5/20	A
exam-07-ch1-6-9	25 Jun 2026	58.5/65	90.0%	12/12	31.5/33	15/20	A+
exam-08-ch1-6-9 (this)	2 Jul 2026	61/65	93.8%	12/12	30.5/33	18.5/20	A+

Strong Section C recovery: 15/20 on exam-07 to 18.5/20 here. Physics subject average: 93.7% (365.5/390, 6 exams). MCQ: 72/72 across all six physics papers (100% all-time).